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Testing Strategy and Acceptance Framework: VI Business (India)

1. Executive Purpose and Document Scope

The Testing Strategy and Acceptance Framework dictates the rigorous, multi-tiered validation approach utilized by VI Business (India) before transferring any managed service (Network, SOC, or Voice) into active production billing.

This artifact proves to the enterprise customer that we do not rely on "plug-and-play" assumptions. Every architectural design is mathematically validated against the signed SLA parameters, ensuring zero operational surprises post-deployment.

2. Definitions, Assumptions, and Applicability

Definitions:

- **SAT:** Site Acceptance Testing.
- **UAT:** User Acceptance Testing.
- **FAT:** Factory Acceptance Testing (Staging/Burn-in).
- **OAT:** Operational Acceptance Testing.

Assumptions:

- Testing criteria are absolute. A site cannot be forcibly migrated into 'Billing Active' status if a P1/P2 defect remains unresolved during the UAT phase.
- Customer IT resources must be available to sign off on specific application performance metrics during the agreed testing window.

Applicability:

- **Target Audience:** Testing Engineers, Pre-Sales SAs, Transition Managers, Bid Writers.
- **Scope:** All new implementations and major change requests (e.g., migrating 500 branches to an SD-WAN overlay).

3. The Testing Streams (The V-Model)

VI Business employs a phased "V-Model" testing lifecycle, ensuring that the physical rollout exactly matches the logical Low-Level Design (LLD).

Testing Phase	Execution Location	Primary Objective	Responsible Party

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Stage 1 (FAT)	Bhiwandi Warehouse	Hardware power burn-in (12 hours). Firmware flash.	Logistics QA Team
Stage 2 (SAT)	Remote Customer Site	Physical installation. Cable labeling. Power stability. Initial ping.	Field Engineer
Stage 3 (UAT)	VI Cloud / Customer DC	Logical routing validation. Application failover testing.	NOC L2 & Customer IT
Stage 4 (OAT)	VI Service Desk	End-to-end ITSM integration. Ensuring alerts auto-generate tickets.	Service Delivery Mgr

4. Entry and Exit Criteria (Quality Gates)

No testing phase may commence without fulfilling "Entry Criteria," and no phase may close without achieving "Exit Criteria."

UAT Example:

- **Entry Criteria:** Customer firewall rules successfully pierced. SAT documentation digitally signed by the customer site manager.
- **Exit Criteria:** 100% of defined critical business applications (e.g., SAP, MS Teams, VoIP) successfully traverse the primary and secondary links within targeted latency/jitter bounds. Zero open P1/P2 defects.

5. Defect Severity Model

If a failure occurs during SAT or UAT, it is categorized mathematically:

- **Severity 1 (Showstopper):** Core routing failure. The site is entirely isolated from the data center. *Action: Deployment halted. Rollback physically initiated.*
- **Severity 2 (Major):** Failover does not trigger properly (e.g., Primary MPLS fails, but Secondary LTE does not engage automatically). *Action: Deployment paused pending immediate NOC L3 remediation.*
- **Severity 3 (Minor/Cosmetic):** The service functions perfectly, but the router hostname is misspelled in the portal, or cable labeling is faint. *Action: Deployment proceeds. Defect logged for remediation within 48 hours.*

6. Performance and Security Testing

- **Performance (Throughput):** VI Business does not rely on simple ICMP pings. UAT utilizes iPerf3 or IxChariot to saturate the link and mathematically prove that the committed bandwidth (e.g., 50 Mbps) is physically available end-to-end without packet loss.
- **Security (Penetration):** Prior to finalizing a bespoke Managed SOC deployment or Cloud Gateway, VI Business contracts an empanelled CERT-In third-party auditor to execute a grey-box penetration test against the newly provisioned IP schema.

7. Acceptance Sign-Off and Billing Trigger

- **The Go-Live Certificate:** Upon meeting all UAT / OAT exit criteria, the VI Transition Manager generates the formal "Service Acceptance Certificate" via Adobe Sign / DocuSign.

- **Deemed Acceptance (Default Clause):** If the customer fails to digitally sign the Acceptance Certificate within three (3) business days of successful UAT completion, and no written technical defects have been formally raised, the site automatically enters "Deemed Acceptance," and SLA/Billing metrics commence.

8. Constraints, Exclusions, and Dependency Notes

- **Third-Party Application Failures:** VI Business's testing boundary ends at network transport. If an iPerf test proves 100 Mbps throughput exists, but the customer's proprietary legacy AS400 application times out, the network is deemed compliant. VI Business assumes no liability for poorly coded legacy application TCP sliding windows.

9. Proposal-Use Guidance

- **RFP Use:** When answering "How do you ensure deployment quality?", quote Section 3 (The V-Model) heavily. It demonstrates a multi-layered defense against human error. Reference Section 6 (iPerf3/IxChariot) when proving bandwidth commitments.

10. Governance and Ownership

- **Owner:** National Head of Quality Assurance & Testing
- **Reviewers:** IT Delivery Director, Deals Desk.
- **Update Cadence:** Annual, aligned with the release of major SD-WAN software branch updates.